

Skills Summary

Interpreting Powers of i and the Complex Plane

Use this reference while completing your worksheet or when studying.

Skill 1 — The cycle of powers of i

$i^2 = -1$, and everything else follows.

i^1	i^2	i^3	i^4	i^5	i^6
i	-1	$-i$	1	i	-1

Because $i^4 = 1$, the powers repeat in blocks of four. **Divide the exponent by 4 and keep only the remainder:** $0 \rightarrow 1$, $1 \rightarrow i$, $2 \rightarrow -1$, $3 \rightarrow -i$.

Example: Simplify i^{34} .

Step 1. The exponent is large, so instead of multiplying it out, locate where 34 lands in the four-step cycle — only the remainder matters.

Step 2. $34 \div 4$ is 8 remainder 2, so $i^{34} = (i^4)^8 \cdot i^2 = 1 \cdot i^2 = -1$.

Try it: Simplify i^{41} .

check:

Skill 2 — Plotting and adding on the plane

The dictionary: $a + bi \longleftrightarrow$ the point (a, b) , with the real part on the horizontal **Re** axis and the imaginary part on the vertical **Im** axis.

Draw each number as an arrow from the origin. Then **adding is tip-to-tail arrow addition**, which in symbols is just adding the parts separately: $(a+bi)+(c+di) = (a+c)+(b+d)i$. The distance from the origin is $|a + bi| = \sqrt{a^2 + b^2}$ — the Pythagorean theorem on the two legs.

Example: $(5 - 2i) + (-3 + 6i)$

Step 1. Addition is the tip-to-tail move on the grid, and that move adds the horizontal steps to each other and the vertical steps to each other.

Step 2. Real parts: $5 + (-3) = 2$. Imaginary parts: $-2 + 6 = 4$. The tip lands at $(2, 4)$, which is $2 + 4i$.

Try it: $(4 + i) + (-6 + 2i)$

check: $-2 + 3i$

Skill 3 — Multiplying by i is a quarter turn

Multiplying by i rotates the arrow 90° counterclockwise about the origin and leaves its length alone.

In symbols: $i(a + bi) = ai + bi^2 = -b + ai$, so the point (a, b) moves to $(-b, a)$ — exactly a quarter turn.

This is *why* the powers of i cycle: four quarter turns is one full turn, so $i^4 = 1$ brings every point home.

Example: $i(2 - 5i)$

Step 1. Multiplying by i should turn the arrow to $(2, -5)$ a quarter turn, so expect the answer in the next quadrant counterclockwise — up and to the right.

Step 2. $i(2 - 5i) = 2i - 5i^2 = 2i + 5 = 5 + 2i$, and indeed $(2, -5)$ has moved to $(5, 2)$.

Try it: $i(-1 + 3i)$

check: $-i - 3$

Skill 4 — Non-real solutions and conjugates

Rule: a quadratic with real coefficients and a negative discriminant has *no* real solutions but exactly two complex ones, and they are always a **conjugate pair**: $p + qi$ and $p - qi$, mirror images across the real axis.

Use $\sqrt{-k} = i\sqrt{k}$ for $k > 0$. On the graph, “no real solutions” just means the parabola misses the x -axis — the solutions live off the real line, not nowhere.

Example: Solve $x^2 - 6x + 25 = 0$.

Step 1. The discriminant $36 - 100$ is negative, so there are no real solutions and the answers will be a conjugate pair — complete the square to see them.

Step 2. $(x - 3)^2 + 16 = 0$, so $(x - 3)^2 = -16$ and $x - 3 = \pm 4i$, giving $3 + 4i$ and $3 - 4i$: the points $(3, 4)$ and $(3, -4)$.

Try it: Solve $x^2 - 2x + 10 = 0$.

check: $1 + 3i$ and $1 - 3i$